Rajiv Gandhi University of Knowledge Technologies- RK Valley-AP

Academic Year 2023– 24: Semester I - Mid Semester Test-II
Civil Engineering - 20CE 2105- Surveying - I
R20 Batch & Regular Mode of Examinations

Year:ENGG-2 | **Date of Examination:December 14, 2023**
Time:45 Minutes | **Max Marks:3X5 = 15**

The Question Paper Consists of 6 questions in 3 sets. Students need to answer 1 questionfrom each set
Each Question Carries 5 Marks and may have 'A', 'B' and 'C' as two sub-questions.

1) Describe the process of repetition and reiteration method of finding out the horizontal angles.

OR

2) Balance the theodolite traverse by Bowditch's rule.

Line	Length	Consecutive Coordinate	
		Latitude	Departure
AB	70.0	+21.500	-65.450
BC	80.0	-80.755	-5.250
CD	43.0	-41.000	+13.550
DE	38.0	-14.250	+35.150
EA	115.0	+114.150	+22.315

3) What is Gale's Table? What is the characteristic of this table?

OR

4) Find the area of the closed traverse having the following data by the coordinate method.

Line	Latitude (y)	Departure (x)
AB	+225.5	+120.5
BC	-245.0	+210.0
CD	-150.5	-110.5
DA	+170.0	-220.0

Rajiv Gandhi University of Knowledge Technologies - RK Valley-AP

Academic Year 2023– 24: Semester I - Mid Semester Test-II
Civil Engineering - 20CE 2105- Surveying - I
R20 Batch & Regular Mode of Examinations

Year: ENGG-2 **Date of Examination: October 18, 2023**

Time: 45 Minutes **Max Marks: 3X5 = 15**

The Question Paper Consists of 6 questions in 3 sets. Students need to answer 1 question from each set

Each Question Carries 5 Marks and may have 'A', 'B' and 'C' as two sub-questions.

1. What are the types of BM that you know of? Explain each type of BM in detail.

OR

2. Compare height of collimation method and rise and fall method.

3. A page of a level book is shown in the following. Fill in the missing readings and calculate the RL of all points. Apply the usual checks.

Station	BS	IS	FS	Rise	Fall	RL	Remar
1	2.150					450.00	BM 1
2	1.645		?	0.500			
3		2.345			?		
4	?		1.965	?			
5	2.050		1.825		0.400		
6	?		?	?		451.50	BM 2
7	1.690		1.570	0.120			
8	2.865		2.100		?		
9			?	?		451.25	BM 3

OR

4. Explain in detail at least four types of Levelling.

5. The following consecutive readings were taken with a dumpy level along a chain line at a common interval of 15m. The first reading was at a chainage of 165m where the RL is 98.085. The instrument was shifted after the fourth and ninth readings.

3.150, 2.245, 1.125, 0.860, 3.125, 2.760, 1.835, 1.470, 1.965, 1.225, 2.390, and 3.035m

Mark rules in the form of a level book page and enter on it the above readings and find the RL of all the points by the Collimation system method.

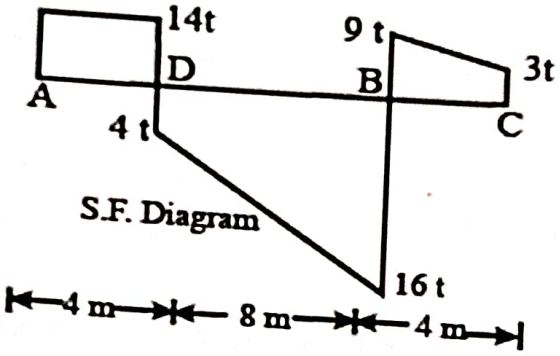
OR

6. The following consecutive readings were taken with a levelling instrument at intervals of 20m.

2.375, 1.730, 0.615, 3.450, 2.835, 2.070, 1.835, 0.985, 0.435, 1.630, 2.255, and -3.630 m.

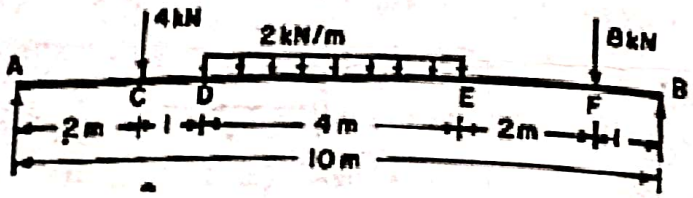
The instrument was shifted after the fourth and eight readings. The last reading was taken on a BM of RL 110.200 m. Find the RLs of all the points by Rise and Fall method.

5. Draw loading diagram from the shear force diagram shown in the figure below(forces are in tonnes and length in meters).



(OR)

6. Calculate support reactions and draw bending moment diagram for the structure shown in the figure below.



Mechanics of Materials – I
Laboratory Examination on 23 December 2023

A. No bending
B. Bending and shear
C. Bending with out shear
D. None of the above

5. A bar of 15mm dia is subjected to a load of 8000N. what will be the stress developed in the bar
   A. 25N/mm²
   B. 35N/mm²
   C. 45N/mm²
   D. None of the above
6. Charpy and izod impact tests are conducted to find the suitability of material for
   A. Gradual loads
   B. Impact loads
   C. Shock loads
   D. None of the above
7. Which type of shear do we test in torsion test
   A. Bending shear
   B. Twisting shear
   C. Both
   D. None of the above
8. Slope at the fixed end of a fixed beam when subjected to udl all along is
   A. Zero
   B. Cannot determine
   C. None of the above
9. Deflection at the fixed end of a fixed beam when subjected to udl all along is
   A. Cannot determine
   B. Zero
   C. None of the above
10. slope at the mid span for a simply supported beam subjected to udl all along is
   A. cannot determine
   B. zero
   C. none of the above

Mechanics of Materials – I
Laboratory Examination on 23 December 2023

Section A(20 marks)

1. Explain in detail the procedure to conduct Charpy Impact test and Izod impact test

(or)

2. Explain in detail the procedure to conduct Double shear test

Section B(20 marks) Answer any 4 out of 5 questions

1. A bolt of 10mm diameter is subjected to double shear. It failed at 8426 Newtons. What is the shear strength of the bolt.
2. What is the reason for specimen to be placed in vertical position in izod impact test and specimen to be placed in horizontal position in charpy impact test.
3. Explain salient points in stress strain curve of mild steel bar
4. A prismatic bar has a cross section of 25mmX50mm and a length of 2meter. Under an axial tensile force of 90kN, the measured elongation of the bar is 1.5mm. compute tensile stress and strain in the bar.
5. Find the minimum diameter of a steel wire, which is used to raise a load of 4000N if the stress in the rod is not to exceed 95MN/m^2

Section C(20 marks)

1. Shear strength of a specimen in double shear test is
   A. (Load)/(2*Area of cross section of specimen)
   B. (Load)/(Area of cross section of specimen)
   C. (Load)/(4*Area of cross section of specimen)
   D. None of the above
2. Brinell hardness number is equal to
   A. P/(spherical area of indentation)
   B. P(perimeter of indentation)
   C. P/(area of indentation)
   D. None of the above
3. Modulus of elasticity is equal to
   A. Stress/(strain)^2
   B. Stress/strain
   C. Strain/stress
   D. None of the above
4. What is meant by pure bending

Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2023 – 24: Semester 1 - Mid Semester Test-II
Civil Engineering - CE2104- Mechanics of Materials I
R20 Batch & Regular Mode of Examinations

Year: ENGG-2 | Date of Examination: 18- Oct -2023
Time: 45 Minutes | Max Marks: 15

3*5=15 Marks

The Question Paper consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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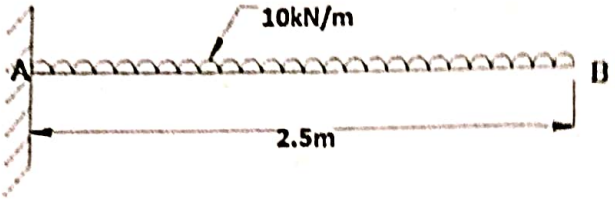
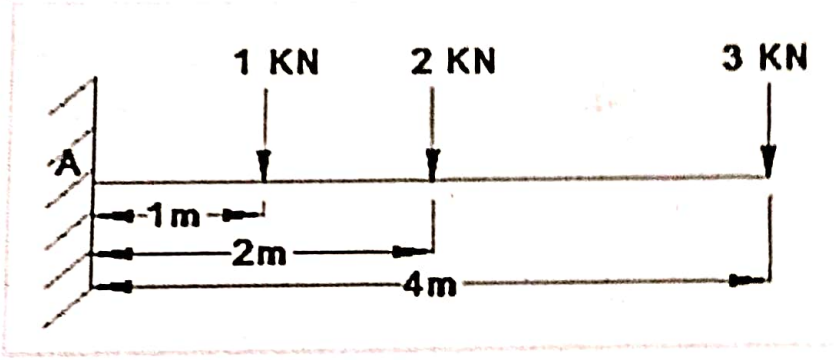
1. Draw free body diagram, calculate support reactions, draw shear force diagram for the structure shown in the figure below.



(OR)

2. Draw free body diagram, calculate support reactions, draw bending moment diagram for the structure shown in the figure below.



3. A. Define shear force,
B. bending moment.
C. what type of stresses will develop in a restrained bar when subjected to reduction in temperature
D. what type of stresses will develop in a restrained bar when subjected to rise in temperature
E. Formula for strain and stress when subjected to temperature change.

(OR)

4. A circular section tapering bar is rigidly fixed at both the ends. The diameter change from 75mm at one end to 150mm at other end, in a length of 1.2m. Compute the maximum stress in the bar if the temperature is raised by 32°C. Take $E = 2X10^5 N/mm^2$. Alpha = $12x10^{-6}$ per °C.

5. A railway is laid so that there is no stress in the rails at 8°C.

A. calculate the stress on the rails at 50°C if there is no allowance for expansion.

B. The stress in the rails at 50°C if there is an expansion allowance of 8mm per rail.

C. The expansion allowance if the stress in the rail is to be zero when the temperature is 50°C.

(OR)

6. Derive expression for elongation of a compound bar subjected to temperature increase. Take steel and copper as compound bar materials.

Rajiv Gandhi University of Knowledge Technologies- RK Valley- AP

Academic Year 2023 – 24: Semester I - Mid Semester Test-2

Civil Engineering - 20CE2103- Mechanics of Fluid

R20 Batch& Regular Mode of Examinations

Year: ENGG-2 | **Date of Examination: 18th October, 2023**

Time: 45 Minutes | **Max Marks: 15 Marks**

3X5 Marks= 15

The Question Paper consists of 6 internal choice Questions. Answer three questions. Each Question Carries 5 Marks

Note: Usage of Scientific calculator is permitted

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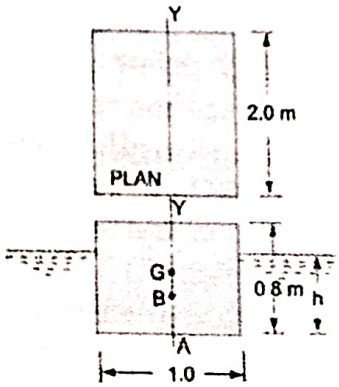
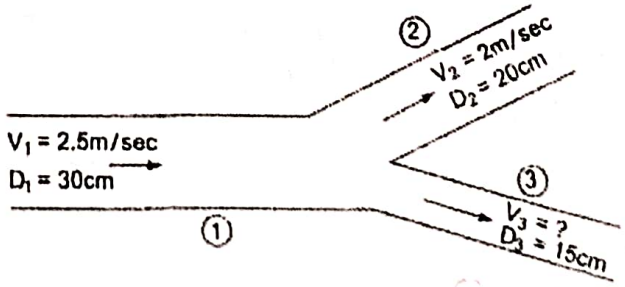
1. A block of wood of specific gravity 0.7 floats in water. Determine the meta-centric height of the block if its size is 2m length x 1m width x 0.8 m depth.



(OR)

2. A rectangular pontoon of 15696 kN weight is floating in water. A weight of 245.25 kN is moved through a distance of 8 m across the deck of pontoon from the center axis, which tilts the ship through an angle of 4^0. Find the meta-centric height of the ship?

3. A 30 cm diameter pipe, conveying water, branches into two pipes of diameter 20 cm and 15 cm respectively. If the average velocity in the 30cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in 15 cm pipe if the average velocity in 20cm diameter pipe is 2 m/s.



(OR)

4. The velocity vector in a fluid flow is given as $V= 4x^3i-10x^2yj+2tk$. Find the velocity and acceleration of fluid particle at (2,1,3) at time t=1.

5. If for a two dimensional flow, the velocity potential is given by $\Phi = x(2y-1)$. Determine the velocity at the point P (4,5). Determine also the value of stream function ψ at the point P. 356.66

(OR)

6. An open circular cylinder of 15 cm diameter and 100 cm long contains water up to a height of 80 cm. Find the maximum speed at which the cylinder is to be rotated about it's vertical axis so that no water spills?

356.66

Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2023 – 24: Semester 1 - Mid Semester Test-2
Civil Engineering - 20CE 2102 – Concrete Technology
R20 Batch & Regular Mode

Year: ENGG-2 | **Date of Examination: 17th Oct 2023**
Time: 45 minutes | **Max Marks: 15**

3X5 Marks= 15

The Question Paper consists of 6 questions with internal choice. Answer any 3 questions from each set. Each Question Carries 5 Marks

Note: Usage of Scientific calculator may be permitted

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1. Explain alkali aggregate reaction and the methods of controlling it.

OR

2. Explain the procedure to determine fineness modulus of fine aggregate using sieve analysis method.

3. Explain the method of determining flakiness index of aggregates.

OR

4. Explain in brief about the following admixtures:
   i) Retarders and pozzolonic materials

5. Explain the method of determining the workability of concrete by compaction factor test.

OR

6. Explain the method of determining setting time of concrete.

Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2023 – 24: Semester I - Mid Semester Test- II

CIVIL Engineering - CE2101-Building Materials And Construction

R20 Batch & Regular Mode of Examinations

Year: ENGG-2 **Date of Examination:17-10-2023**

Time: 45Minutes **Max Marks: 15**

3X5 Marks= 15

The Question Paper consists of 3 units. Answer one full question from each unit. Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

1) Write down the difference between clamp burning and kiln burning?

(OR)

2) Explain in detail various types of glass used in the construction?

3) Explain in detail the manufacturing process of ceramics?

(OR)

4) Explain in detail

a) Deval's attrition test of stone? Why this experiment is used? (3M)

b) Smith's test of stone

5) Explain in detail the types of mortar used for construction?

(OR)

6) Explain in detail the characteristics of plastics used in construction?